

Volatility forecasting and risk signal analysis

VIX forecasts, uncertainty and relative risk regimes

A comparison of short-horizon VIX forecasts alongside a separate monitoring signal that classifies current volatility relative to its recent history.

Methods	R; ARIMA; regression; rolling evaluation; lagged percentile thresholds
Data	FRED VIX series; 2 January 1990–21 May 2026
Revision	21 September 2026

1 Executive summary

This project combines VIX forecasting with a separate signal that classifies current volatility relative to recent history. VIX is an implied volatility index, so the project is forecasting the index itself rather than realised volatility. The daily VIX sample runs from 1990-01-02 to 2026-05-21 and contains 9,191 clean observations. A rolling backtest was then run over 775 one-day-ahead forecasts.

The regression model had the lowest VIX-level RMSE at 1.7805, compared with 1.7865 for ARIMA. The difference is small, so the result should be read as a marginal improvement rather than a clear win. The interval coverage was close to the 95% target for both models: 94.58% for ARIMA and 94.84% for the regression model.

The residual checks gave a more cautious view. The ARIMA Ljung-Box p-value was 0.7791, while the regression p-value was 0.0106. This suggests the regression model was slightly better on point forecast accuracy, but the ARIMA residuals were cleaner from an autocorrelation perspective.

The risk signal splits the VIX history into three states using rolling thresholds based only on past data. Normal days made up 74.47% of the 8,939 observations with available thresholds, Elevated days 12.73%, and Stress days 12.80%. Stress days had an average VIX of 27.93 and a maximum VIX of 82.69, describing the high-volatility observations selected by this relative threshold rule. This is not evidence that the signal predicts future stress.

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2 Data and scope

The project uses daily closing values of the Cboe Volatility Index, commonly known as VIX. The data were downloaded from FRED using the VIXCLS series. FRED reports this as a daily, close, not seasonally adjusted index series sourced from Cboe [1]. Cboe describes the VIX index as a measure of expected 30-day volatility implied by S&P 500 option prices [2].

The data were cleaned by removing missing observations and retaining strictly positive VIX values. A log transformation was used for the forecasting models, since VIX is positive and can spike sharply during market stress.

Table 1. Data summary.

Metric	Value
Data start	1990-01-02
Data end	2026-05-21
Clean observations	9191
Rolling forecasts	775
Risk signal observations	8939

3 Methodology

3.1 Forecasting models

Two forecasting models were used.

- **ARIMA model:** an automatic ARIMA specification was selected using an information criterion. This follows the standard approach implemented in the R `forecast` package, which is described by Hyndman and Khandakar [3].
- **Regression model:** a regression model was selected using AIC over lagged and rolling VIX features. The features included lagged log-VIX, short and medium moving averages, and previous risk-regime indicators. AIC is a standard model selection criterion introduced by Akaike [4].

The main evaluation used rolling one-day-ahead forecasts. This is more realistic than fitting a model once on the full sample, because the forecast at each date only uses information available before that date.

3.2 Risk signals

The risk signal is based on rolling VIX thresholds. For each date, the model compares the current VIX value with the 75th and 90th percentiles of the previous 252 trading days. This gives three regimes:

- **Normal:** VIX is below the elevated threshold.

- **Elevated:** VIX is above the 75th percentile threshold but below the stress threshold.
- **Stress:** VIX is above the 90th percentile threshold.

The thresholds are lagged, so the signal avoids look-ahead bias. This matters because the purpose is to produce a monitoring signal that could have been calculated at the time.

3.3 Back-transformation and forecast dates

The models forecast log-VIX. Exponentiating the log forecast gives a central forecast on the VIX scale; under a Gaussian log forecast distribution it is a median, not a bias-adjusted arithmetic mean. Prediction-interval bounds are exponentiated in the same way.

The final 20-step forecast is labelled with successive weekdays generated by the supplied script. An exchange-holiday calendar is not applied, so these dates should be read as illustrative weekday labels rather than a verified schedule of trading sessions.

4 Results and discussion

4.1 Forecast accuracy

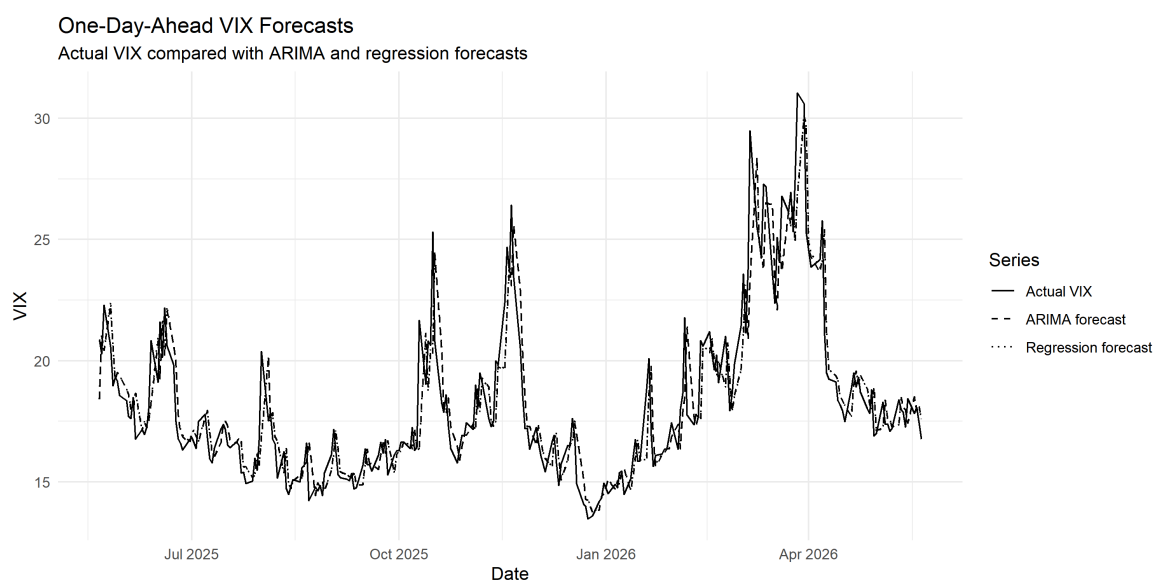


Figure 1. One-day-ahead VIX forecasts. The ARIMA and regression forecasts both track the broad movement in VIX, although sharp spikes remain difficult to anticipate.

The two forecasting models produced very similar point forecast accuracy. The regression model had a slightly lower RMSE on the VIX level, but ARIMA was very close. Both models also had 95% interval coverage close to the nominal target.

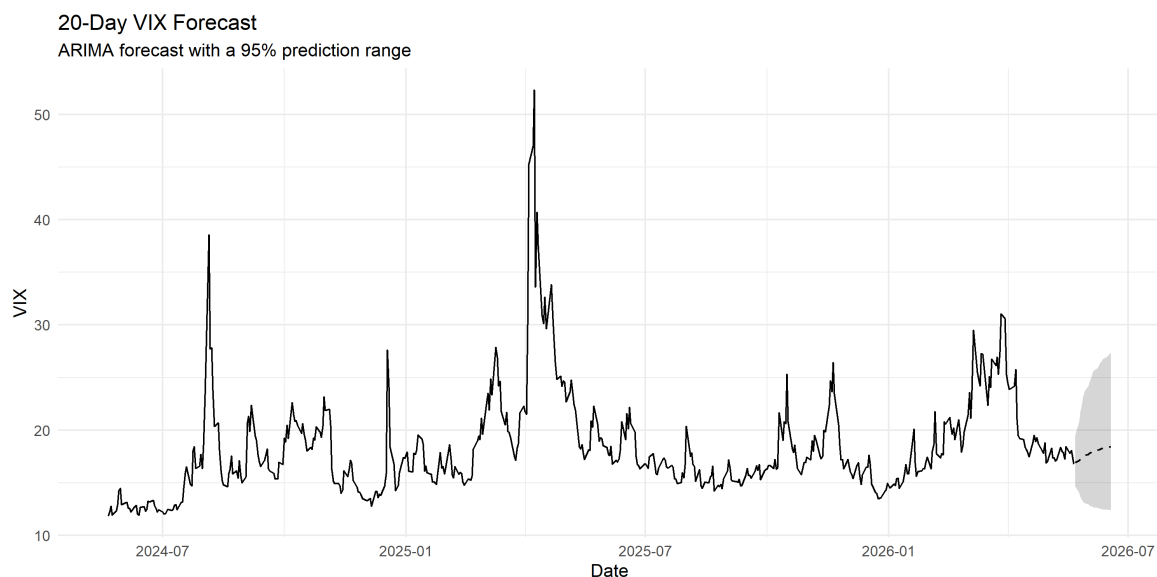
The small gap between the two RMSE values means the regression result should not be overstated. In practical terms, both models perform similarly on short-horizon VIX forecasts.

Table 2. Model comparison over 775 rolling one-day-ahead forecasts.

Metric	ARIMA	Regression
RMSE, log scale	0.0766	0.0765
MAE, log scale	0.0514	0.052
RMSE, VIX level	1.79	1.78
MAE, VIX level	0.9822	0.9937
MAPE, VIX level (%)	5.09	5.17
95% coverage, log scale (%)	94.58	94.84
95% coverage, VIX level (%)	94.58	94.84
Forecast comparisons	775	775

VIX-level RMSE is rounded in the table; the reported unrounded values are 1.7865 for ARIMA and 1.7805 for regression.

4.2 Forecast uncertainty

**Figure 2.** Final 20-step ARIMA forecast. The shaded region shows the 95% prediction range.

The final 20-step forecast is relatively flat, with a mild upward drift in the central forecast and a widening prediction range. This is expected: uncertainty grows as the forecast horizon extends. The interval is more informative than the point forecast because VIX can move sharply during stress periods.

Table 3. Final 20-step VIX forecast, selected columns.

Date	Central VIX	Lower 95%	Upper 95%
2026-05-22	16.94	14.66	19.57
2026-05-25	17.10	14.06	20.79
2026-05-26	17.25	13.68	21.74
2026-05-27	17.38	13.42	22.50
2026-05-28	17.50	13.23	23.15
2026-05-29	17.61	13.08	23.71
2026-06-01	17.71	12.96	24.20
2026-06-02	17.80	12.87	24.63
2026-06-03	17.89	12.79	25.00
2026-06-04	17.96	12.73	25.34
2026-06-05	18.03	12.68	25.64
2026-06-08	18.09	12.63	25.91
2026-06-09	18.15	12.59	26.16
2026-06-10	18.20	12.56	26.38
2026-06-11	18.25	12.52	26.59
2026-06-12	18.29	12.49	26.77
2026-06-15	18.33	12.47	26.94
2026-06-16	18.36	12.44	27.10
2026-06-17	18.39	12.42	27.24
2026-06-18	18.42	12.40	27.37

4.3 Residual diagnostics

Residual diagnostics were used to check whether the models left behind strong autocorrelation. The Ljung-Box test is a standard portmanteau test for remaining autocorrelation in time-series residuals [5].

Table 4. Ljung–Box residual tests at lag 20.

Model	Statistic	p-value	Interpretation at the 5% level
ARIMA	11.48	0.7791	No strong evidence of residual autocorrelation.
Regression	28.95	0.0106	Residual autocorrelation remains.

The diagnostic trade-off is clear: regression has marginally lower point-forecast RMSE, while ARIMA has less evidence of remaining residual autocorrelation. The ranking depends on which aspect of model performance is being assessed.

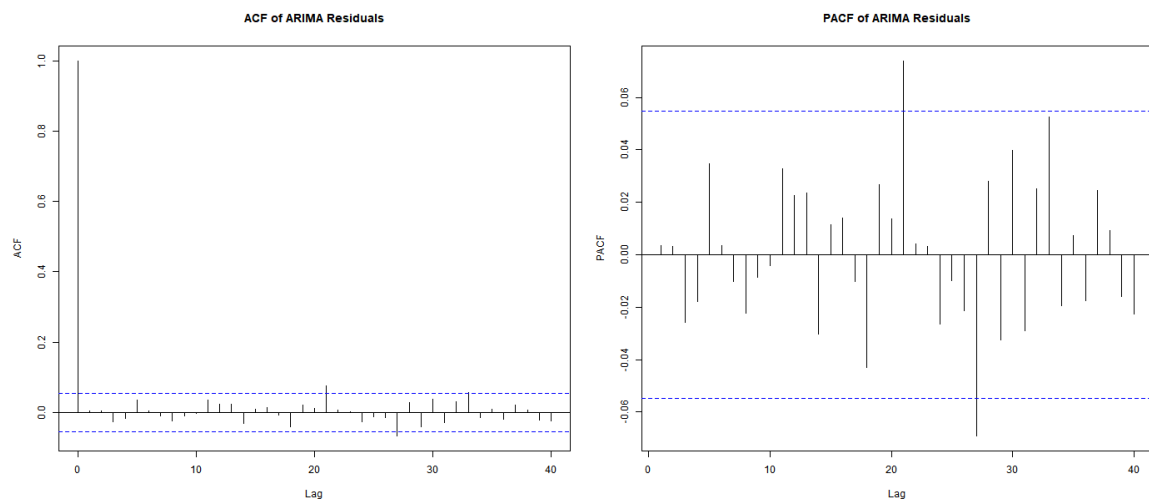


Figure 3. ARIMA residual ACF and PACF. The residuals show limited remaining autocorrelation, consistent with the Ljung-Box result.

4.4 Risk regimes

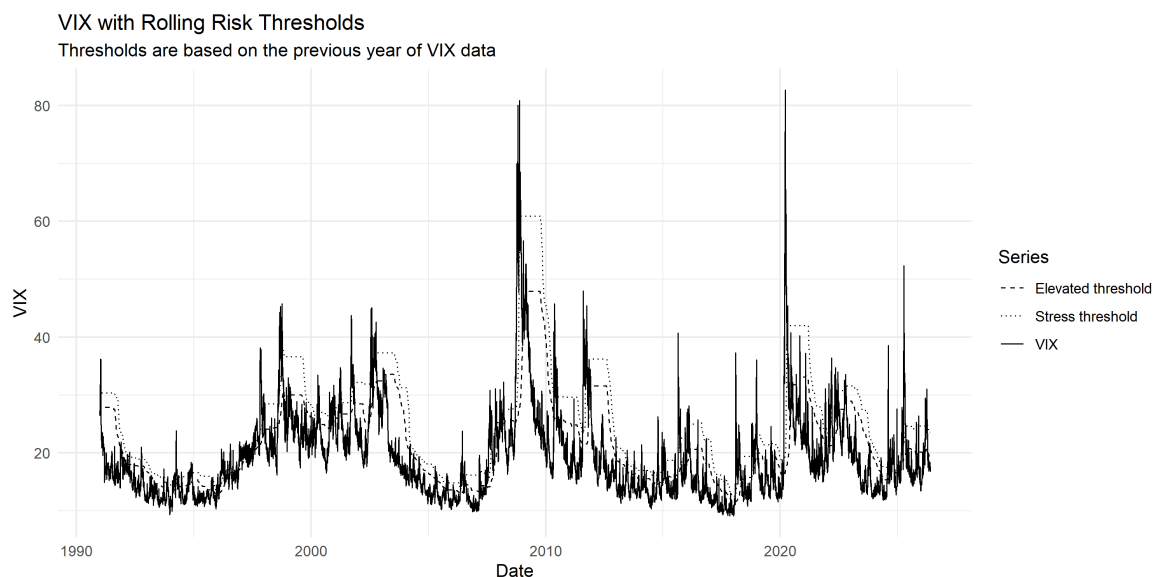


Figure 4. VIX with rolling risk thresholds. The elevated and stress thresholds move over time because they are based on the previous year of VIX data.

The rolling thresholds adapt to the volatility environment. This is useful because a VIX value that is high in a calm market may not be unusual during a persistent high-volatility period. The signal therefore measures risk relative to recent conditions, not against a fixed VIX level.

Table 5. Overall risk-regime summary.

Regime	Days	Share (%)	Avg VIX	Median VIX	Max VIX	Avg z-score
Stress	1144	12.80	27.93	25.47	82.69	2.39
Elevated	1138	12.73	22.87	20.69	59.93	0.91
Normal	6657	74.47	17.28	16.11	46.35	-0.62

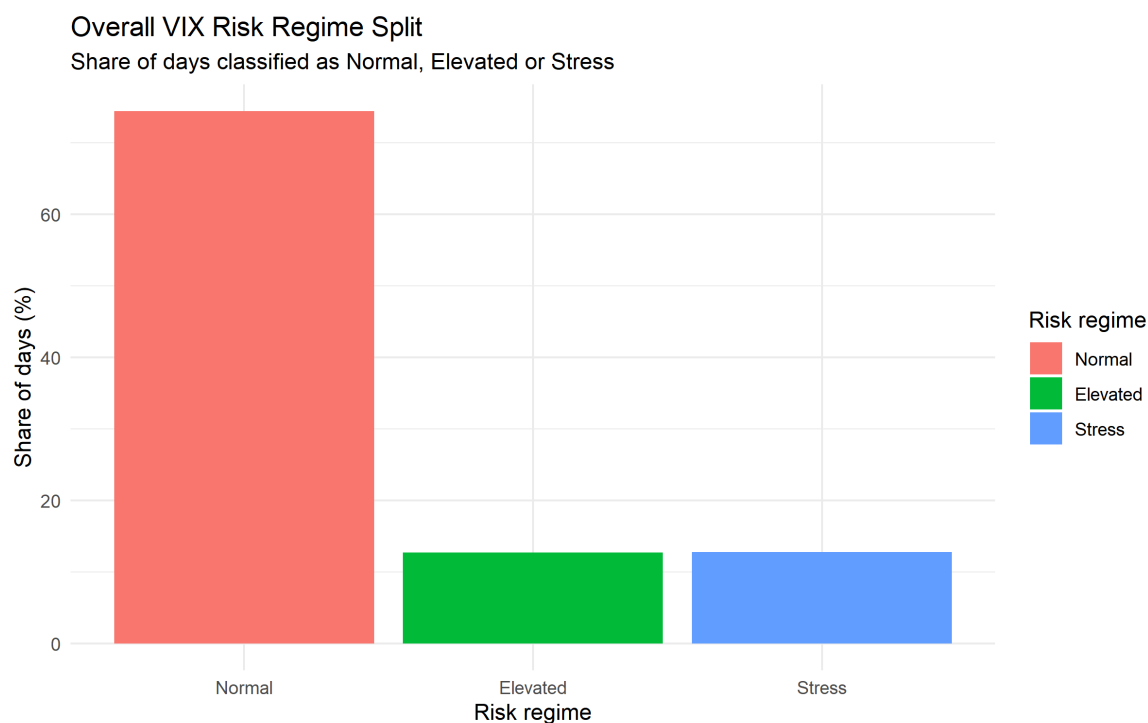


Figure 5. Overall split of Normal, Elevated and Stress regimes.

Normal periods make up most of the observations with available thresholds, while Elevated and Stress regimes are much smaller but have clearly higher average VIX values. The stress regime has an average VIX of 27.93 and includes the largest observed VIX spike in the sample.

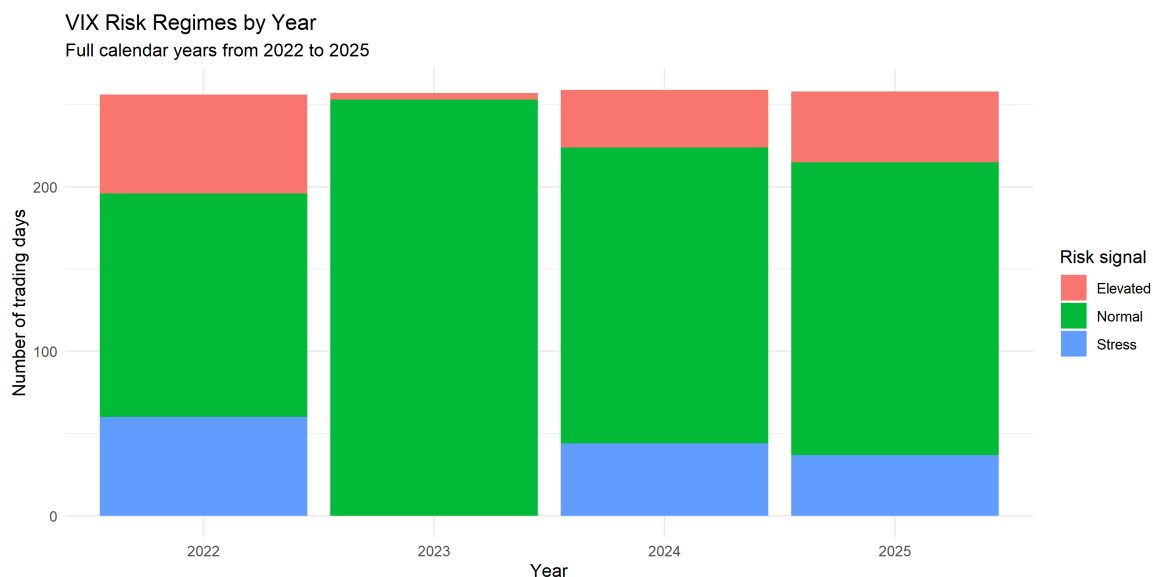


Figure 6. Risk regimes by year for full calendar years 2022 to 2025.

The yearly view shows that 2022 had a large number of Elevated and Stress days, while 2023 was almost entirely Normal. Stress periods reappeared in 2024 and 2025. Because the thresholds are rolling, the regime label is relative to the recent volatility environment.

Table 6. Risk-regime days by year.

Year	Normal	Elevated	Stress
2022	136	60	60
2023	253	4	0
2024	180	35	44
2025	178	43	37

4.5 Final observed signal

At the final date in the sample, the VIX was 16.76 and the signal was **Normal**. The elevated threshold was 19.19 and the stress threshold was 23.56. Since the observed VIX was below both thresholds, the final observation was classified as Normal.

Table 7. Latest risk signal.

Date	VIX	Signal	z-score	Elevated threshold	Stress threshold
2026-05-21	16.76	Normal	-0.43	19.19	23.56

5 Limitations and further work

The main limitation is that the project forecasts the VIX index itself. VIX is an implied volatility measure, so the analysis should not be described as a direct realised-volatility forecast. A useful extension would be to compare VIX forecasts with realised S&P 500 volatility or to add a simple persistence benchmark.

The small RMSE difference has not been subjected to a formal forecast-comparison test. Regression residual autocorrelation also limits a simple interpretation of its model-based prediction intervals.

Risk labels are contemporaneous descriptions relative to the previous 252 observations. They do not establish a trading strategy, predict future crises or measure absolute risk. Future work could evaluate whether the labels add information about subsequent returns or realised volatility.

The final forecast uses direct exponentiation and weekday date labels. A production version should distinguish medians from means and apply the relevant exchange calendar.

6 Conclusion

The project produces a coherent VIX forecasting and monitoring workflow. The regression model achieved the lowest RMSE in the rolling backtest, but the improvement over ARIMA was small. ARIMA produced cleaner residual diagnostics, so it remains a useful benchmark and a sensible model for the final forecast.

The risk signal is the most interpretable output from the project. It classifies VIX conditions into Normal, Elevated and Stress regimes using thresholds that are based only on past data. This gives a simple framework for monitoring market conditions without relying on fixed VIX cut-offs.

References

- [1] Federal Reserve Bank of St. Louis. *CBOE Volatility Index: VIX (VIXCLS)*. FRED economic data series. Available at: <https://fred.stlouisfed.org/series/VIXCLS>.
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